STAGE1:Generative COT Enhancement Engine

step1 VQA Alignment

{"Q": "Has this leaf contracted Early Blight?",
"A": "No, this tomato leaf has Late Blight."}

Processing method:images resize to **384x384**; QA extract **18.93%** preserving original distribution;Shuffled into **single sequence**

step2 Automatic CoT Generation.

Prompt Template:

Type detection (Diagnosis/Control)

Few-shot examples (3 demos)

Step-by-step guidelines

Hyperparams: T=0.3, max_tokens=1024

Deepseek-VL2

step3 CoT Judge and Enhenced

GPT-4

RESPONSE:<think>Step 1: Analyze disease characteristics... Step 2: Recommend cultural/preventive practices ... **</think><answer>**No, this tomato leaf has Late Blight**</answer>**

Guideline:
Accuracy ;Completeness ;LogicalFlow ;Relevance ;
Clarity Threshold: **τ = 8.0/10.0**

STAGE2: GRPO Reinforcement Learning

step1 Domain Vocabulary Construction

Plant Synonyms
15 species x 5-8 variants

Disease Synonyms
20 diseases x 3-6 variants

Treatment Keywords
4 categories, 47 terms

5-tier fuzzy matching
(1.0 → 0.25)

Step2 GRPO Training Engine

Dynamic Dual-Path Evaluation

Diagnosis	Treatment
Plant+ Disease	4 Categories

Qwen-2.5VL-3B

System Prompt:
Requirements:Output as steps (Step 1, 2, 3, 4).Use only one step set per question type. Never mix.
Step Guidelines（For Identification Questions）：
Step 1: Identify host plant.Step 2: Describe symptoms. Step 3: Analyze disease pattern.Step 4: Provide diagnosis & confidence.
Output Format： **<think>**(Step reasoning) **</think><answer>**(Final Answer) **</answer>**

Step3 Reward &Update

Total Reward: R(o) = 0.5·R_f + 2.0·R_a + 0.5·R_r [0, 3.0]

Format R_f	Answer R_a (Main)	Reasoning R_r
Structure:<think><answer>	Diagnosis:	Logical coherence (0.25)
Step count: 3-4 steps	Plant match (0.8)	Professionalism (0.15)
Step quality (length)	Disease match (1.2)	Completeness (0.10)
Reasonable length	Control:	
	Treatment keywords (2.0)	

Group Relative Advantage:
num_generations = **3**(OOM prevention) ; advantage =(r-μ)/ σ ; Relative evaluation

Clipped PPo Update:
max grad norm = **0.3**(gradient explosion prevention) ; KL penalty + stability ;
DeepSpeed **ZeRO-3** (accelerate, balance); BF16 + grad_accum=4